

SSN College of Engineering, Kalavakkam
Department of Computer Science and Engineering
V Semester - CSE 'A'
UCS2511 NETWORKS LAB
Exercise 2
Echo Client

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Aim:

Echo Client Message Over Network

```
#include <stdio.h>
#include <netinet/in.h>
#include <sys/socket.h>
#include <sys/types.h>
#include <string.h>
#include <stdlib.h>
#include <unistd.h>

int main(int argc, char **argv)
{

    int socketID, portNum;
    struct sockaddr_in serverAddr;

    socketID = socket(AF_INET, SOCK_STREAM, 0); // IPv4 TCP/UDP IP

    serverAddr.sin_family = AF_INET;
    portNum = atoi(argv[1]);
    serverAddr.sin_port = htons(portNum);
    serverAddr.sin_addr.s_addr = INADDR_ANY;

    //printf("Server : serversoc %d\n", socketID);
```

```

    bind(socketID, (struct sockaddr*) &serverAddr,
sizeof(serverAddr));

listen(socketID, 10);
int clientSocket = accept(socketID, NULL, NULL);

//printf("Server : clientsoc %d\n", clientSocket);
char data[255] = " ";
for(int i=2; i< argc; i++)
{

    strcat(data,argv[i]);

    strcat(data, " ");
}

send(clientSocket, data, sizeof(data),0);
recv(clientSocket, data, sizeof(data),0);

printf("Echoed Message from client: %s", data);

close(clientSocket);
close(socketID);

return 0; }

```

client.c

```

#include <stdio.h>
#include <netinet/in.h>
#include <sys/socket.h>
#include <sys/types.h>
#include <string.h>
#include <stdlib.h>
#include <unistd.h>

```

```
int main(int argc, char **argv)
{

    int socketID, portNum;
    struct sockaddr_in serverAddr;


    socketID = socket(AF_INET,SOCK_STREAM,0); // IPv4 TCP/UDP IP


    serverAddr.sin_family = AF_INET;
    portNum = atoi(argv[1]);
    serverAddr.sin_port = htons(portNum);
    serverAddr.sin_addr.s_addr = INADDR_ANY;


    //printf("Client : %d\n",socketID);


    connect(socketID, (struct sockaddr*) &serverAddr,
sizeof(serverAddr));


    char data[255];
    recv(socketID, data, sizeof(data),0);


    printf("Message from server: %s", data);

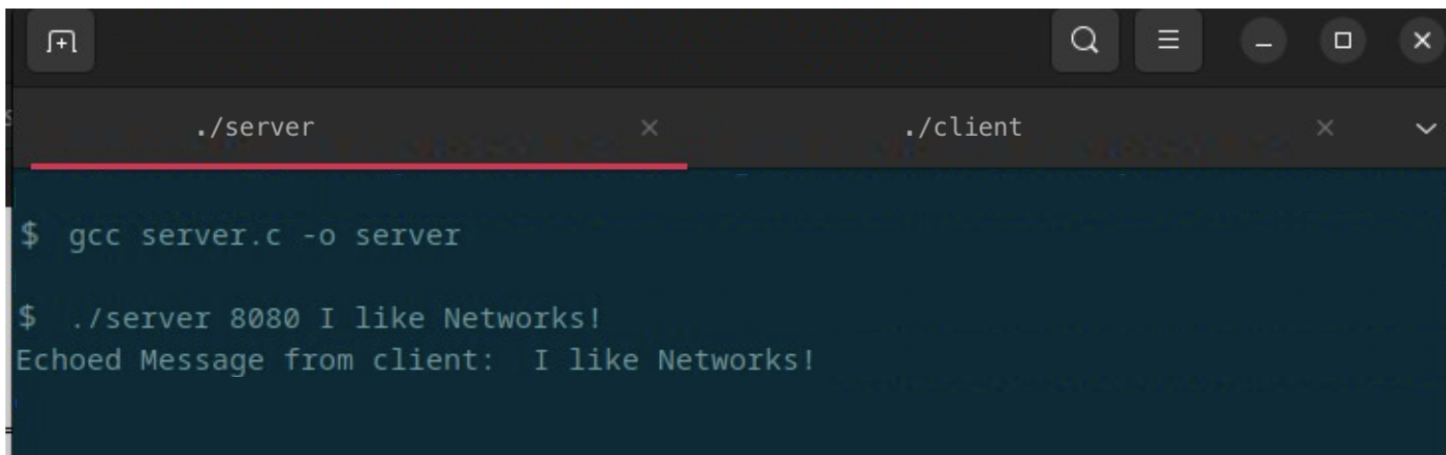

    //sendto(socketID, data, sizeof(data),0,(struct
sockaddr*) &serverAddr, sizeof(serverAddr));


    send(socketID, data, sizeof(data),0);
    close(socketID);
    return 0;

}
```

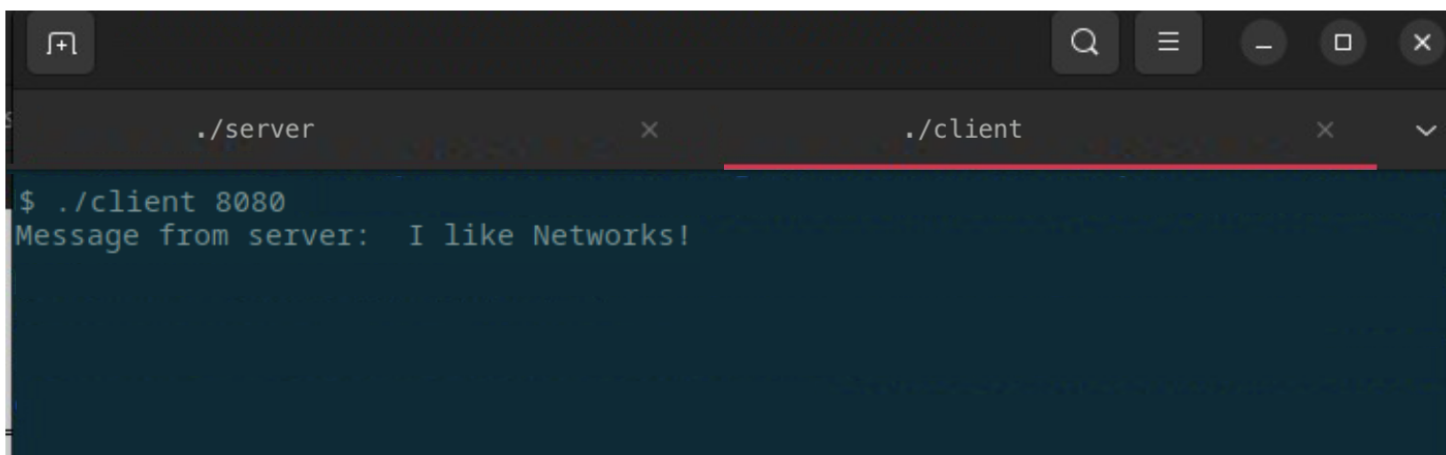
Output:

O/P:



A terminal window with two tabs: `./server` (active) and `./client`. The terminal shows the following commands and output:

```
$ gcc server.c -o server
$ ./server 8080 I like Networks!
Echoed Message from client: I like Networks!
```



A terminal window with two tabs: `./server` and `./client` (active). The terminal shows the following commands and output:

```
$ ./client 8080
Message from server: I like Networks!
```

Learning Outcomes:

I learned how to echo text using TCP.

